

KRISHNA KousHIK THOKALA

Austin, TX (Open to Relocation) | (317) 840-3830 | krishnakoushikthokala@gmail.com | LinkedIn | GitHub | Portfolio

Summary

AWS-certified Solutions Architect and MS in Computer Science with hands-on experience building AWS and Kubernetes infrastructure using Terraform, GitOps, and CI/CD, with EKS delivery, observability, and autoscaling. Backed by 2 years of 24/7 carrier-grade incident response — leading 5-engineer shifts through triage, RCA, and SLA-driven restoration for 25+ Tier-1 clients.

Certifications

AWS Certified Solutions Architect – Associate | Amazon Web Services | 2026

Technical Skills

Cloud Infrastructure: AWS, VPC, IAM, EKS, ECR, RDS, Lambda, DynamoDB, ALB, Terraform (IaC)

Kubernetes & Delivery: Kubernetes, Docker, Helm, Argo CD, GitOps, GitHub Actions, CI/CD

Reliability & Observability: CloudWatch, Prometheus, Grafana, HPA, k6, on-call incident response, RCA, SLOs, MTTR

Programming & Systems: Python, Bash, SQL, PostgreSQL, FastAPI, REST APIs, Linux

Networking: TCP/IP, BGP, MPLS, DNS, Security Groups, NetworkPolicies

Projects

CloudOps GitOps Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions

GitHub

- Built a GitOps platform on EKS where 4 Argo CD Applications reconcile Git-defined state across isolated dev/staging/prod and observability namespaces, delivering automated drift self-healing and one-commit rollback recovery.
- Provisioned the AWS foundation (VPC, EKS, ECR, IAM) with Terraform and secured delivery with GitHub OIDC, RBAC, and immutable commit-SHA promotion from dev to prod.

CloudOps SRE Platform | EKS, Terraform, Helm, FastAPI, React, RDS

GitHub

- Built an SRE operations dashboard on EKS that centralizes service health, deployments, incident timelines, and MTTR into a single view, on a FastAPI/React/PostgreSQL stack backed by RDS.
- Packaged the service with Helm (health probes, resource limits, ALB Ingress) and hardened it with Secrets Manager credentials, restricted RDS access, and structured JSON logging.
- Validated HPA autoscaling under k6 load, scaling backend replicas from 2 to 6 pods to serve 2,035 requests at 99.5% success and 1.52s p95 latency.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

GitHub

- Built a serverless uptime monitor on AWS where EventBridge triggers scheduled Lambda checks, DynamoDB stores status history, SNS sends state-change alerts, and a CloudFront React dashboard shows live state.
- Designed a latest-status access pattern that cut current-status reads from 86,400 rows to 10 per lookup, eliminating full-table scans as history grows.
- Ran a controlled outage drill validating 1-second failure detection, DOWN-state dashboard visibility within 26 seconds, and alert deduplication that suppressed repeat-DOWN spam while confirming recovery.

Experience

Tata Communications

July 2022 – July 2024

Senior Engineer, Network Operations Center – Shift Lead

Pune, India

- Led 5-engineer shifts in a 24/7 on-call NOC, owning a queue of 25+ tickets per shift across 25–30 Tier-1 carrier clients.
- Drove high-severity incident response end to end – triage, fault isolation, escalation, and restoration – resolving complex failures to root cause through structured log, alarm, and segment-level analysis.
- Sustained 99.9% service availability and a 4-hour restoration SLA across the client base by prioritizing P1/P2 incidents, driving timely resolution, and maintaining proactive customer communication.
- Authored post-incident RCAs and structured shift handoffs to drive recurrence prevention and continuity; earned a Certificate of Excellence for resolving an escalated incident into a documented customer win.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10

Chennai, India